ITP105

NETWORKING & COMMUNICATION 2

Week 4:

INTER-VLAN ROUTING

Introducing Inter-VLAN Routing

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What is Inter-VLAN Routing?

- Each VLAN is a **unique broadcast domain**
  - Computers on separate VLANs are, by default, not able to communicate.
- Each VLAN is a **unique IP subnetwork**.

- To allow VLANs to communicate, we need a router to communicate among separate broadcast domains and unique IP subnetworks.

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What is Inter-VLAN Routing?

- **Inter-VLAN Routing** is the process of enabling communication between devices on different VLANs within the same network.

Inter-VLAN Routing

Methods:

- Traditional Inter-VLAN Routing
- Router-on-a-stick Inter-VLAN Routing
- Switch Based Inter-VLAN Routing

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Traditional Inter-VLAN Routing:

- Also called as Legacy Inter-VLAN Routing.
- Requires multiple physical interfaces on both the router and switch
  - Router needs an interface for each VLAN and the switch needs a switchport for each VLAN to be routed
  - Switchports connecting to router are in ACCESS mode for the VLAN travelling up the link

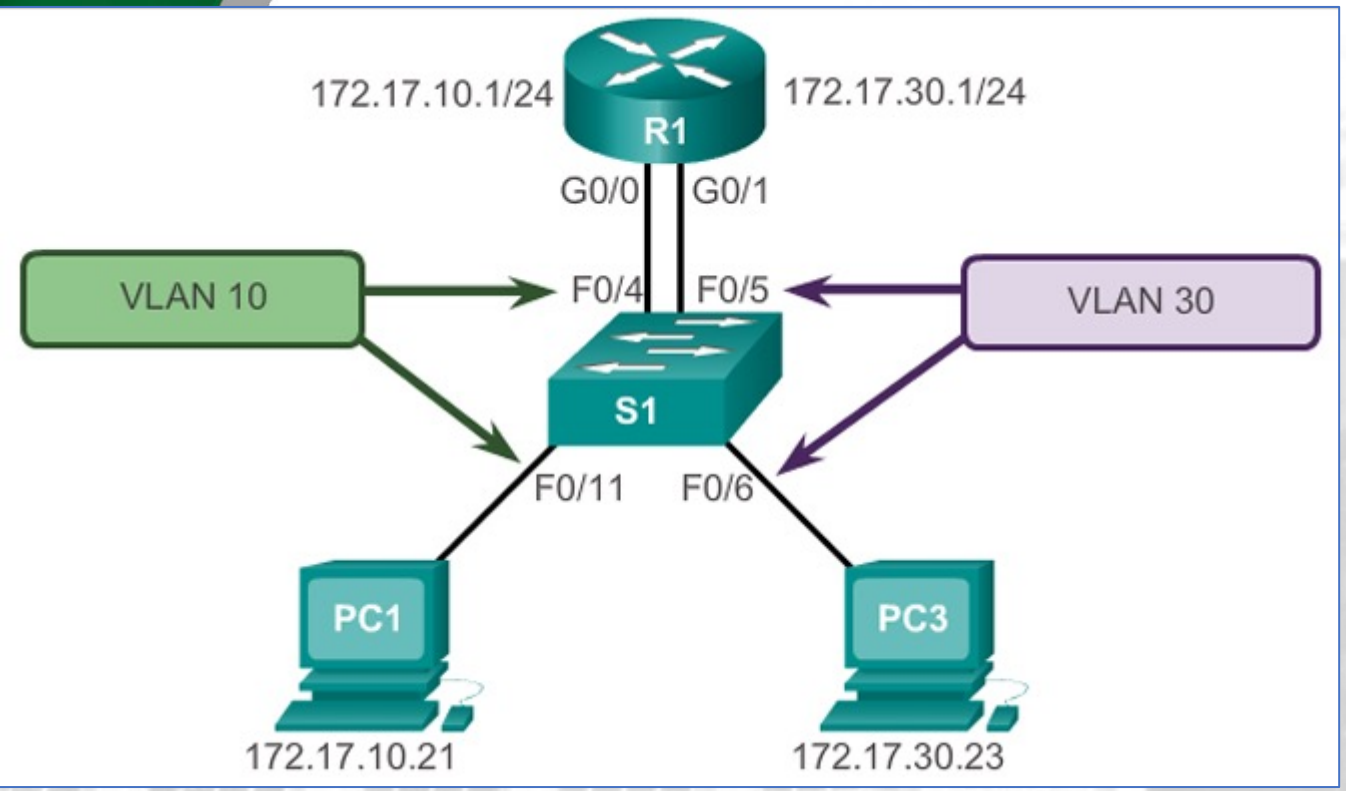
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Traditional Inter-VLAN Routing:

- Routers have a limited number of physical interfaces.
- Switches require a switchport for each VLAN as well
  - Uses up available switchport on the switch
  - More links makes the cabling more complex
- Each VLAN has a dedicated link to the router
  - As the link is not shared with multiple VLANs throughout is better and performance is

Traditional Inter-VLAN Routing:

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Switch Configuration

```
S1(config)#vlan 10
S1(config-vlan)#vlan 30
S1(config-vlan)#interface f0/11
S1(config-if)#switchport access vlan 10
S1(config-if)#interface f0/4
S1(config-if)#switchport access vlan 10
S1(config-if)#interface f0/6
S1(config-if)#switchport access vlan 30
S1(config-if)#interface f0/5
S1(config-if)#switchport access vlan 30
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

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Traditional Inter-VLAN Routing:

```
R1(config)#interface g0/0
R1(config-if)#ip address 172.17.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#interface g0/1
R1(config-if)#ip address 172.17.30.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Inter-VLAN Routing

Traditional Inter-VLAN Routing:

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.17.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.17.10.0/24 is directly connected, GigabitEthernet0/0
L       172.17.10.1/32 is directly connected, GigabitEthernet0/0
C       172.17.30.0/24 is directly connected, GigabitEthernet0/1
L       172.17.30.1/32 is directly connected, GigabitEthernet0/1
```

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Router-on-a-stick Inter-VLAN Routing:

- A single physical interface used to route all VLANs
- The physical interface is divided into subinterfaces
  - Each VLAN assigned to a separate subinterface
  - Each subinterface is configured to trunk using dot1q
  - Switchport (to router) is configured as a trunk link
  - Messages from all VLANs are tagged and

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Router-on-a-stick Inter-VLAN Routing:

- Only one physical interface is used on router and one trunk port is used on the switch
- All VLANs use the single trunk so there is potential for a bottle neck resulting in reduced performance

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Router-on-a-stick Inter-VLAN Routing:

WHAT ARE SUBINTERFACES?

- Subinterfaces are virtual interfaces which are associated with one physical interface
  - Each subinterface has an IP and a VLAN
  - Each subinterface is the gateway for the VLAN
  - Each subinterface retags the VLAN traffic with the destination VLAN ID as the traffic enters the trunk

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Router-on-a-stick Inter-VLAN Routing:

CONFIGURING SUBINTERFACES:

- Similar to configuring physical interfaces
  - Create the subinterface
  - Assign it to a VLAN
  - Assign an IP address
  - Enable interface

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Router-on-a-stick Inter-VLAN Routing:

CREATE THE SUBINTERFACE/S:

- The syntax for the subinterface is always the physical interface, followed by a period and a subinterface number
- The subinterface number is configurable, but it is typically associated to reflect the VLAN number

R1(config)# `interface [interface].nn`

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Router-on-a-stick Inter-VLAN Routing:

ASSIGN THE SUBINTERFACE/S TO A VLAN:

- Before assigning an IP Address, the interface must to be configured to operate on a specific VLAN using the proper encapsulation

R1(config)# `encapsulation dot1q` *`vlan-id`*

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Router-on-a-stick Inter-VLAN Routing:

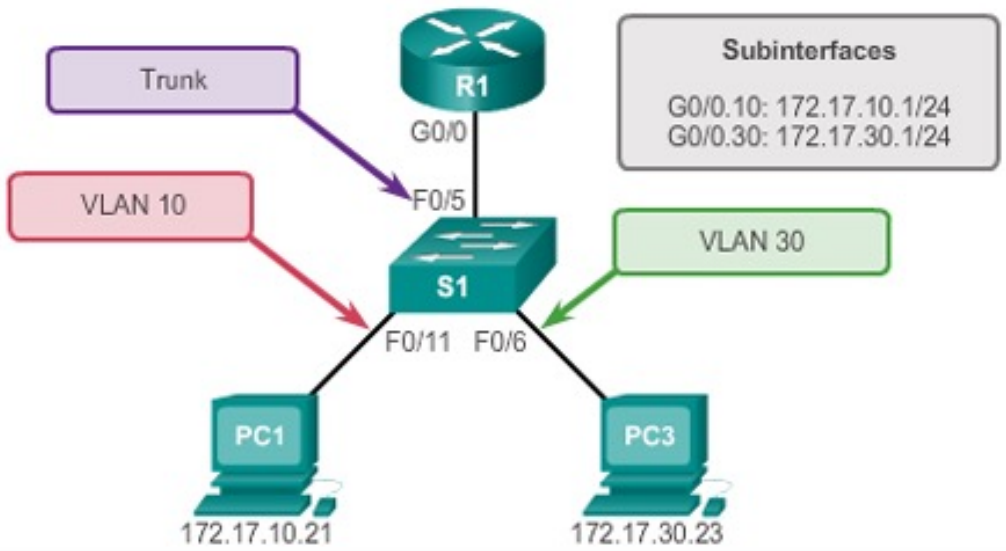
ASSIGN AN IP ADDRESS:

- The IP Address assigned here will become the default gateway for that VLAN

```
R1(config-subif)# ip address [address] [mask]
```

Inter-VLAN Routing

Router-on-a-stick Inter-VLAN Routing:




```
S1(config)# vlan 10
S1(config-vlan)# vlan 30
S1(config-vlan)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1#
```

```
R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 172.17.10.1 255.255.255.0
```

VLAN 10

```
R1(config-subif)# interface g0/0.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# ip address 172.17.30.1 255.255.255.0
```

VLAN 30

```
R1(config)# interface g0/0
R1(config-if)# no shutdown
*Mar 20 00:20:59.299: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
 changed state to down
*Mar 20 00:21:02.919: %LINK-3-UPDOWN: Interfa
 changed state to up
*Mar 20 00:21:03.919: %LINEPROTO-5-UPDOWN: Line protocol on
 changed state to down
*Mar 20 00:21:02.919: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
 changed state to up
*Mar 20 00:21:03.919: %LINEPROTO-5-UPDOWN: Line protocol on
 Interface GigabitEthernet0/0, changed state to up
```

Enable interfaces

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Router-on-a-stick Inter-VLAN Routing:

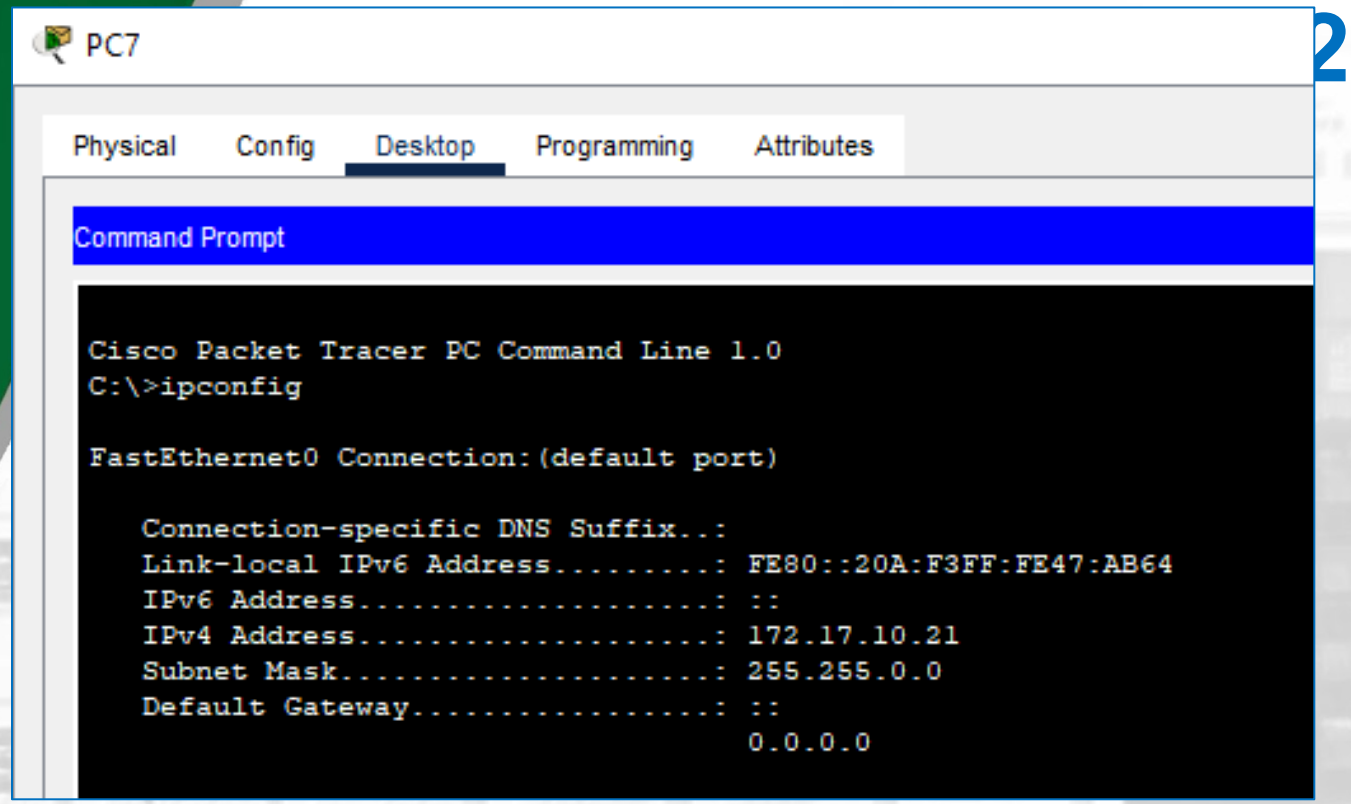
Verifying Routing

- Access to devices on remote VLANs can be tested using the ping command
- The ping command sends an ICMP echo request to the destination address
- When the host receives an ICMP echo request, it responds with an ICMP echo reply
- Tracert is a useful utility for confirming the routed path taken between two devices

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Router-on-a-stick Inter-VLAN Routing:

Verifying Connectivity Between PC1 to 2

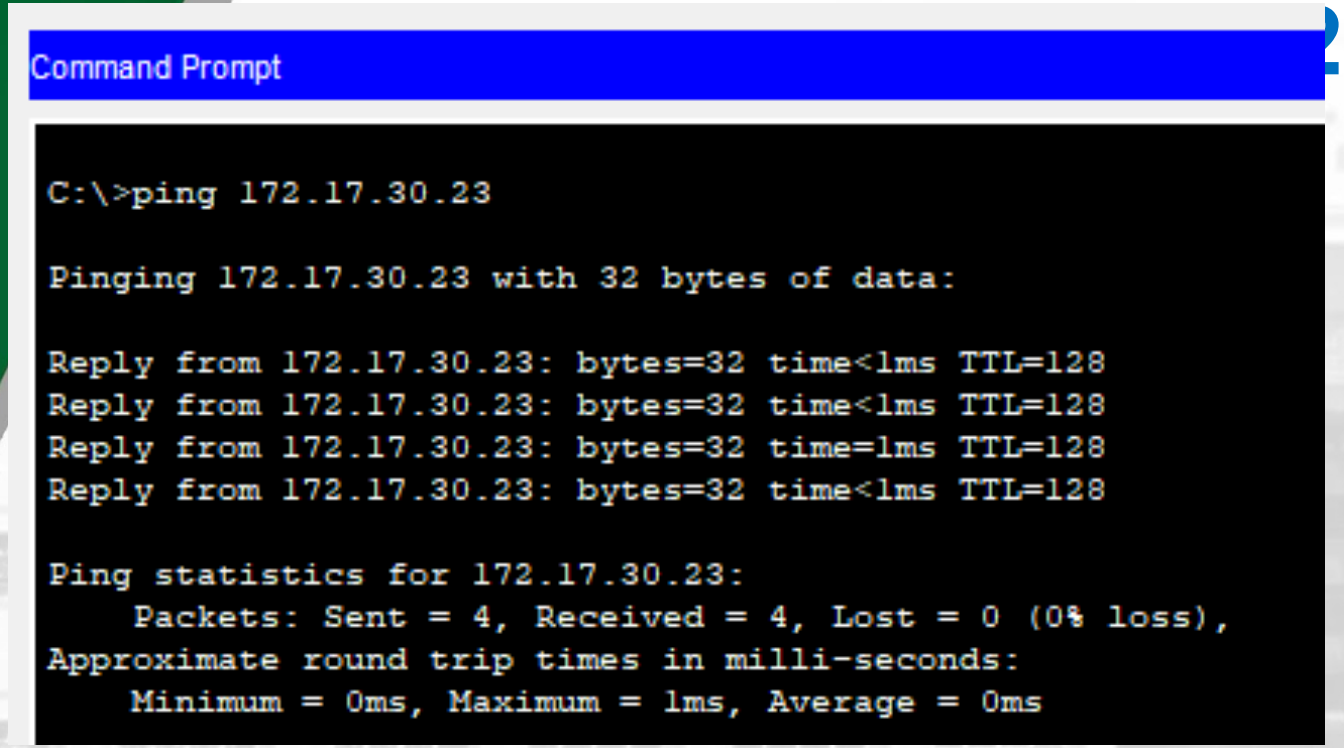


- From a host, verify connectivity to a host in another VLAN using the **ping** command. It is a good idea to first verify the current host IP configuration using the **ipconfig** Windows host command.
- The output confirms the IPv4 address and default gateway of PC1

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Router-on-a-stick Inter-VLAN Routing:

Verifying Connectivity Between PC1 to



- Use **ping** to verify connectivity with PC2 and S1.
- The **ping** output successfully confirms that inter-VLAN routing is operating.

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Router-on-a-stick Inter-VLAN Routing:

Verifying Subinterfaces

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
       B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF,
       IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
            type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1,
       L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default,
       U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
       l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

   172.17.0.0/16 is variably subnetted, 4 subnets, 2 masks
C     172.17.10.0/24 is directly connected, GigabitEthernet0/0.10
L     172.17.10.1/32 is directly connected, GigabitEthernet0/0.10
C     172.17.30.0/24 is directly connected, GigabitEthernet0/0.30
L     172.17.30.1/32 is directly connected, GigabitEthernet0/0.30
```

```
R2>enable
R2#show ip route | begin Gateway
Gateway of last resort is not set

     172.17.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.17.10.0/24 is directly connected, GigabitEthernet0/0.10
L       172.17.10.1/32 is directly connected, GigabitEthernet0/0.10
C       172.17.30.0/24 is directly connected, GigabitEthernet0/0.30
L       172.17.30.1/32 is directly connected, GigabitEthernet0/0.30
```

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Router-on-a-stick Inter-VLAN Routing:

Verify Subinterface IP Address and

```
R2#show int g0/0.10
GigabitEthernet0/0.10 is up, line protocol is up (connected)
  Hardware is PQUICC_FEC, address is 0007.ec8c.4b01 (bia 0007.ec8c.4b01)
  Internet address is 172.17.10.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation 802.1Q Virtual LAN, Vlan ID 10
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last clearing of "show interface" counters never

R2#show int g0/0.30
GigabitEthernet0/0.30 is up, line protocol is up (connected)
  Hardware is PQUICC_FEC, address is 0007.ec8c.4b01 (bia 0007.ec8c.4b01)
  Internet address is 172.17.30.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation 802.1Q Virtual LAN, Vlan ID 30
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last clearing of "show interface" counters never
```

This command displays detailed information about the sub-interface GigabitEthernet 0/0.10 on a Cisco router. This sub-interface is typically used in a Router-on-a-Stick configuration for inter-VLAN routing

Specifies the type of hardware.

Indicates VLAN tagging is used

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Router-on-a-stick Inter-VLAN Routing:

Verify Subinterface IP Address and Status

- The **show ip interface brief | include up** command is typically used to filter the output of the **show interface brief** command to only display interfaces that are in the "up" state. The output would show interfaces that are operationally up and administratively up.

```
R1#show ip interface brief | include up
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.99  192.168.99.1    YES manual up                    up
R1#
```

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Router-on-a-stick Inter-VLAN Routing:

Verifying Trunk Link Status

```
S2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1
Fa0/6       desirable    n-802.1q       trunking      1
Fa0/11      desirable    n-802.1q       trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-1005
Fa0/6       1-1005
Fa0/11      1-1005

Port        Vlans allowed and active in management domain
Fa0/5       1,10,30
Fa0/6       1,10,30
Fa0/11      1,10,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,30
Fa0/6       1,10,30
Fa0/11      1,10,30
```

The show interfaces trunk command is used on a Cisco switch to display the trunking status and configuration of all trunk interfaces

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Router-on-a-stick Inter-VLAN Routing:

Verifying Switch

```
S1#show interfaces f0/3 switchport
Name: Fa0/3
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More--
```

The configured mode of the interface is **trunk**, which means it is set to carry traffic for multiple VLANs.

The access VLAN configured for the port (*if it were in access mode*) is VLAN 1, which is the default VLAN for many switches.

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Router-on-a-stick Inter-VLAN Routing:

Router Interface and Subinterface

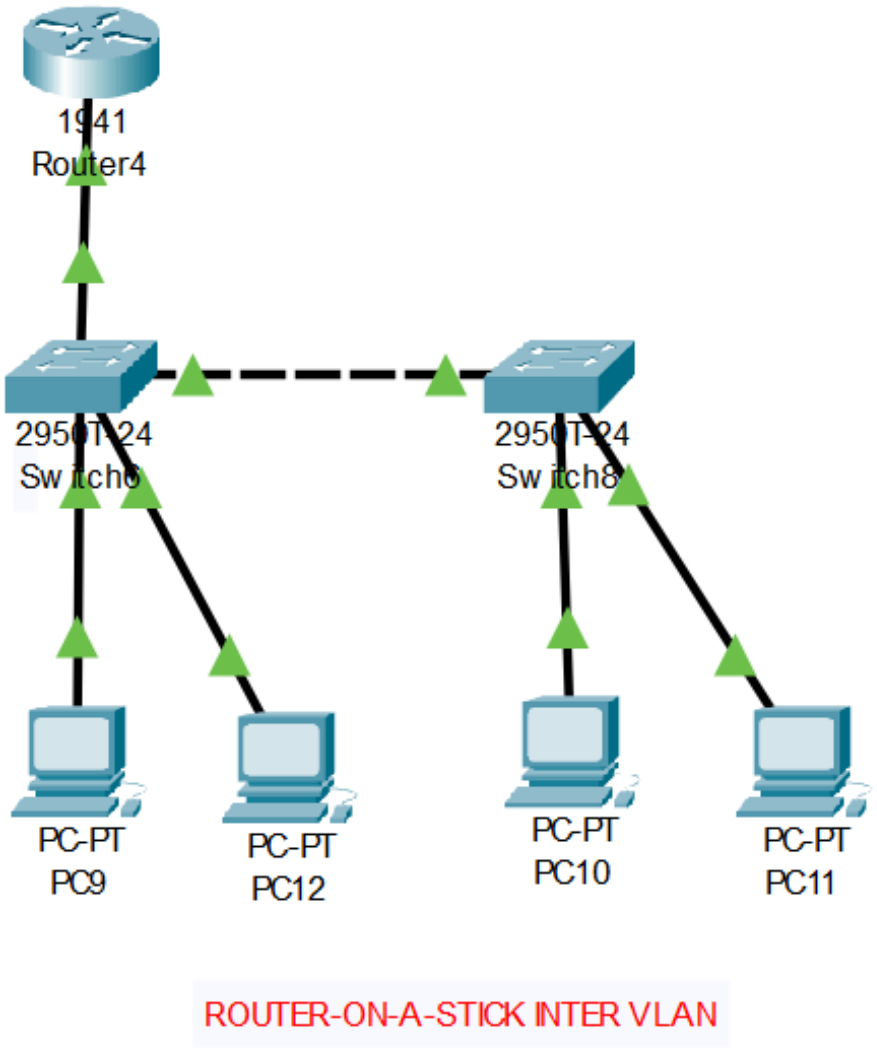
Physical Interface	Subinterface
One physical interface per VLAN	One physical interface for many VLANs
No bandwidth contention	Bandwidth contention
Connected to access mode switch port	Connected to trunk mode switch port
More expensive	Less expensive
Less complex connection configuration	More complex connection configuration

Any questions?

Inter-VLAN Routing

MIDTERM LAB ACTIVITY #2:

Router Interface and Subinterface Comparison



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ACTIVITY:

Router Interface and Subinterface Comparison

OBJECTIVE: Set up inter-VLAN routing using a Router-on-a-Stick configuration with a Cisco router and a switch, involving multiple VLANs.

Requirements

Devices:

- 1 Cisco Router (e.g., 1941)
- 1 Cisco Switch (e.g., 2960)
- 4 PCs (for different VLANs)

VLANs:

- VLAN 10 (Sales)
- VLAN 20 (Engineering)
- VLAN 30 (HR)
- VLAN 40 (Finance)

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ACTIVITY:

Router Interface and Subinterface Comparison

Steps *(Provide screenshot of CLI commands/configurations):*

1. Create VLANs on the switch
2. Assign Ports to VLANs
3. Configure trunking on the Switch
4. Configure the Router
5. Configure PCs
6. Test Connectivity

Thank you for listening!
God bless!